M.I.N.E.R.S.

Mobile Intelligent Navigation and Emergency Rescue System

2. Problem Statement

Underground mine rescue teams face unknown hazards, poor visibility, difficult terrain, survivor-location uncertainty and unreliable communication, making initial human inspection dangerous and slow.

3. Solution proposed

M.I.N.E.R.S. is a low-cost AI-integrated rescue rover that:

- Monitors CH_4, CO, O_2 and environmental hazards.
- Uses 360° 2D LiDAR for mapping, navigation and obstacle detection.
- Uses one stereo camera for both RGB vision and depth estimation.
- Uses thermal + RGB + depth + posture/motion analysis for survivor detection.
- Builds a dynamic rescue map showing tunnels, hazards, survivors and communication status.
- Uses risk-aware navigation considering hazards, communication and battery level.
- Supports bounded autonomous operation during temporary communication loss.
- Stores mission data locally and synchronizes it after reconnection.
- Provides a live rescue dashboard with video, thermal data, hazards, survivor alerts and rover status.
- Includes an independent Safety Governor, manual override and emergency stop.

4. Implementation Plan

Build and integrate the rover in phases:

Sensors & Drive → ROS 2 → SLAM/Navigation → AI Perception → Risk & Mission Planning → Communication Resilience → Rescue Dashboard → Testing

Target beneficiaries:

Mine rescue teams, underground workers, mine safety officers and emergency response teams.

Expected outcomes:

Safer reconnaissance, faster hazard assessment, improved survivor localization and better rescue-team situational awareness.

5. **Safety and Efficiency Analysis**

M.I.N.E.R.S. reduces the need for immediate human entry into potentially hazardous areas by performing initial reconnaissance remotely.

Safety:

- Multimodal sensing and sensor redundancy
- Obstacle avoidance and localization monitoring
- Communication-loss safe behaviour
- Battery and fault monitoring
- Independent Safety Governor
- Manual override and emergency stop

Efficiency:

- Rapid underground reconnaissance
- Real-time hazard mapping
- Faster survivor identification
- Centralized rescue information through the dashboard

6. Scalability and Future Development

The modular architecture allows future upgrades such as:

- Industrial-grade communication and relay networks
- Improved thermal and gas sensors
- UWB-assisted localization
- Advanced edge AI and mine-specific datasets
- Multi-rover cooperative exploration
- Improved structural-risk analysis
- Certified mine-deployable hardware
- Adaptation for tunnels, disaster zones and confined-space inspection

7. Innovation Advantage

M.I.N.E.R.S. does not rely on a single AI model or sensor. Its key innovation is the integration of:

Multimodal Survivor Intelligence

+

Risk-Aware Navigation

+

Communication-Resilient Bounded Autonomy

+

Dynamic Rescue Intelligence Map

+

Independent Safety Governor

During communication loss, the rover can continue a predefined bounded mission, store information locally and attempt safe recovery/reconnection instead of immediately losing the mission.

This creates a unified rescue-support system rather than a conventional inspection rover with isolated sensing capabilities.

8. Visual Impact & User Journey

BEFORE:

Mine Emergency

→ Unknown Tunnel Conditions

→ Human Entry for Reconnaissance

→ Hazard Exposure

→ Uncertain Survivor Location

→ Limited Situational Information

WITH M.I.N.E.R.S.

Mine Emergency

→ Rover Deployment

→ Autonomous/Assisted Exploration

→ Hazard Mapping

→ Thermal + RGB + Depth Survivor Detection

→ Risk Assessment

→ Communication-Resilient Operation

→ Live Rescue Intelligence Dashboard

→ Informed Human Rescue Decision

DETAILED PROJECT REPORT

Full architecture, components, AI pipeline, communication system, implementation roadmap, testing methodology, research references and future development:

LINK FOR DETAILED REPORT-
https://docs.google.com/document/d/1ss40h_uYsy6qUH0NHCVeRIjkne2ntDT1jjmRSOYxkq8/edit?usp=sharing